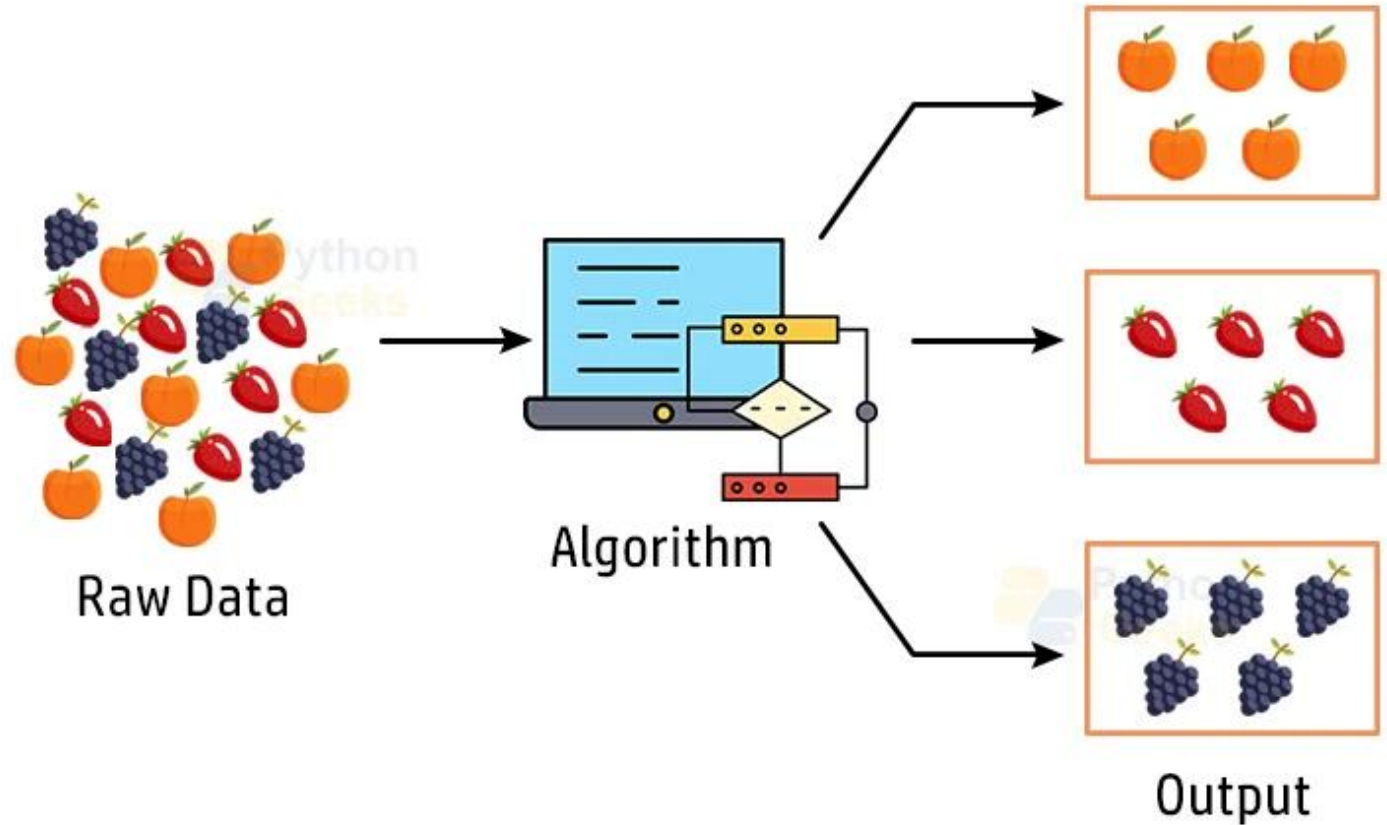


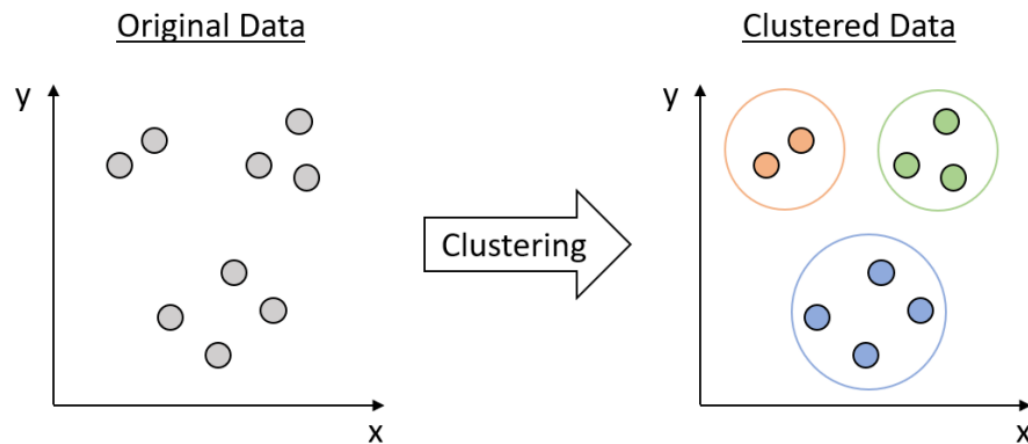
Unsupervised Learning (Clustering)

FINDING PATTERNS IN THE
UNKNOWN

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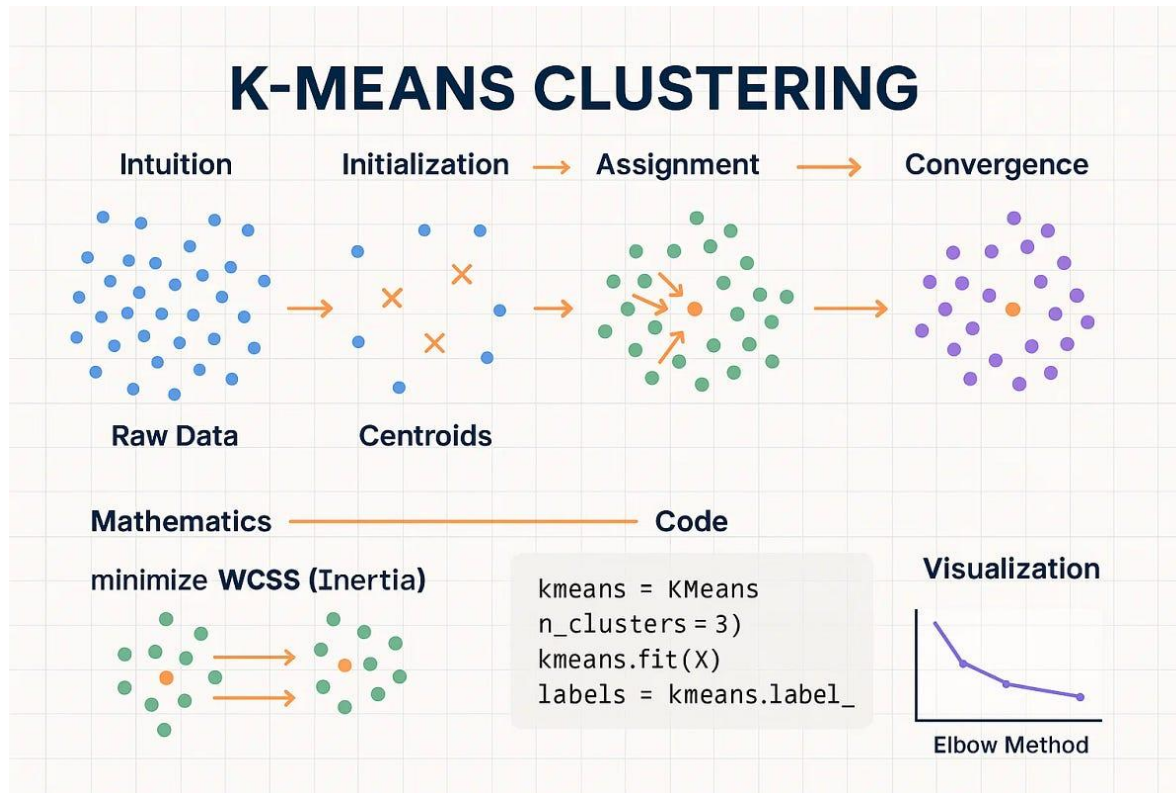


What is Clustering?



- **The Concept:** Giving an algorithm data with no labels and asking it to group similar items together.
- **The Goal:** Discovering hidden structures or natural groupings that a human might miss.
- **Analogy:** Sorting a pile of mixed coins into groups based on their size and weight without knowing the value of each coin.

K-Means Clustering

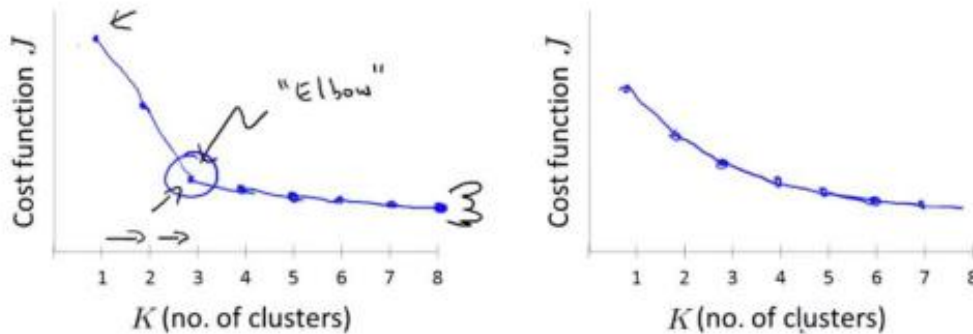


- **Definition:** An algorithm that splits a dataset into a specific number (K) of distinct groups.
- **How it Works:**
 - The user chooses a value for K.
 - The algorithm places K "centroids" in the data.
 - Points are assigned to the nearest centroid, and centroids move until the groups are stable.
- **Real-World Use:** Grouping store customers into tiers (e.g., Bronze, Silver, Gold) based on spending habits.

The Elbow Method

Choosing the value of K

Elbow method:



Source: Ng, no date

- **The Problem:** How do we know if we should split our data into 2, 3, or 10 groups?
- **The Solution:** A visual chart used to figure out exactly how many clusters (K) the data should be split into.
- **The Logic:** We look for the "elbow" point on the graph where adding more clusters no longer significantly improves the grouping.